Brady Bangasser

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Systems engineer focused on reliability and platform work: running production HPC and cloud infrastructure, automating operations, and keeping large-scale systems available and observable under real load.

EXPERIENCE

System Administrator, Iowa State University Apr 2025 – Present

Ames, IA

- Operate and maintain a production HPC cluster with SLURM, Kubernetes, Active Directory, Ansible, Prometheus, and Apptainer, keeping it secure and available for a large user base.
- Built diagnostic and monitoring workflows that isolate hardware degradation, network bottlenecks, and software-stack faults before they take nodes offline.
- Built a TUI stress-testing tool in C that dynamically schedules HPL (LINPACK) jobs and watches live metrics to keep runs inside safe thermal and power limits, protecting node health while pushing for peak results.
- Automated node provisioning and recovery with Ansible so failed or new nodes rejoin service quickly.
- Built Prometheus-based monitoring and alerting to catch cluster issues before users notice.

Head of Networking, Cardinal Space Mining Aug 2024 – Present

Ames, IA

- Designed, built, and tested the internal network for a competition mining robot, ensuring redundant control and telemetry throughout each arena run.
- Collected multiple camera streams and relayed them to a third-party platform for live judging.
- Hardened networking equipment against wireless interference, dust, and physical damage in the competition arena.

Cloud Security Engineer, John Deere May 2025 – Aug 2025

Johnston, IA

- Built and deployed tooling that verified 100,000+ AWS and Azure resources against company policy, NIST standards, and platform best practices.
- Designed and managed least-privilege IAM policies in AWS to tighten cloud access control.
- Wrote automated remediation for common misconfigurations flagged across the cloud estate.
- Led a migration of IAM, security, and dev-management tooling from Python to Go, a ~10x performance gain that cut AWS Lambda runtime costs by roughly the same factor.
- Deployed high-throughput Go services on AWS EKS to run compliance checks across the cloud estate.

Lead Telemetry Engineer, Cyclone Rocketry Aug 2024 – May 2025

Ames, IA

- Built modular software to control and acquire data from airframe instruments, improving reliability and maintainability.
- Built a ground-station TUI in C to visualize live telemetry from the airframe during flights.

Research Assistant — High Performance Computing, Bethel University Aug 2023 – Aug 2024

Saint Paul, MN

- Built and configured two bare-metal HPC clusters from scratch: networking, custom DNS and routing, user management, and SLURM job scheduling with VMware and Debian.
- Developed a self-healing IoT sensor mesh network to detect earthquakes, landslides, and flash floods in remote regions.

PROJECTS

Erid — Unified Cloud Images 2026

- One Packer + Terraform pipeline that builds identical, reproducible qcow2 golden images across every major cloud (AWS, GCP, Azure, Oracle, Hetzner) and on-prem, applying the same hardening everywhere to eliminate drift.

- Standardized a hardening baseline applied identically to every target image, so security posture no longer varies by platform.
- Produces multiple image variants for different node roles, plus matching container images, all from the same definition.

Tech: Terraform, Packer, AWS, GCP, Azure, Oracle, Hetzner, qcow2, Linux

Omni HTTP Router 2025–2026
- Wrote a compiler that lowers a routing DSL to native object files and generates matching Terraform and binaries for easy, fast cloud deployment.
- Built a Rust HTTP router that resolves middleware at compile time into fully static stacks, routing a request in under 30 instructions on SIMD-enabled x86.

Tech: Rust, LLVM, Terraform, gRPC

Computer Vision Parking Enforcement Radar 2025–2026
- Architected a Redis Streams pipeline for low-latency, high-throughput image and detection routing from a REST API to processing servers, backed by PostgreSQL geospatial storage.
- Tuned the broker for low-latency, high-throughput delivery from the REST API to processing servers.

Tech: Python, Redis Streams, PostgreSQL, REST

Public Data Scraping and Processing 2025
- Built a parallel, scalable pipeline to scrape, clean, and store public datasets, deployed via a Terraform CI/CD pipeline to AWS and a personal Kubernetes cluster.
- Wrote a compiler-like tool that pairs lexical analysis with an LLM to standardize and validate data formats, and trained an ML model to flag anomalies in the collected data.

Tech: C++, C, Python, Lua, Rust, MySQL, Docker, PyTorch, Terraform, Kubernetes

Gin Gonic Routing & Middleware Framework 2025–2026
- Wrote a Rust tool that auto-generates and protects Go (Gin) routes by file path, with middleware for permission bits, request IDs, and structured logging.
- Generated per-route middleware for permission bits, request IDs, and structured logging.

Tech: Rust, Go, Gin Gonic

EDUCATION

M.S. Computer Science, Iowa State University Aug 2025 – May 2027

GPA: 4.00
- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

B.S. Computer Science, Iowa State University Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List · Minors in Cybersecurity and Data Science
- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, TypeScript
Infrastructure & Platform: Kubernetes, Docker, Terraform, Packer, Ansible, SLURM, Prometheus, Apptainer, Linux/Unix, Git, CI/CD
Cloud: AWS, Azure, Google Cloud, AWS EKS
Data & Backends: PostgreSQL, Redis
Security: NIST standards, Least-privilege IAM